# GIOVANNI SULLUTRONE

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**I do stuff that I like, pretty much.**

If you want my unprofessional opinion on what I do, I simply keep up with papers, get excited, and try to break or find new stuff. The papers you see here all started either from discussions with researchers or users, or from something in the literature that I vehemently disagreed with (or the opposite).

## EDUCATION

**PhD, Information and Communication Technologies** November 2023 - October 2026
Research on LLM evaluation, with a focus on refusal behaviour, agent interaction, and applications in digital libraries. Expected completion: October 31, 2026.

**M.Sc. in Computer Engineering** October 2021 - October 2023
Artificial Intelligence and Intelligent Robotics track
University of Salerno, Graduated *110/110 cum laude*

**B.Sc. in Computer Engineering** October 2018 - October 2021
University of Salerno, Graduated *110/110*

## RESEARCH EXPERIENCE

**PhD Researcher - LLM** Modena, Italy
*University of Modena and Reggio Emilia* November 2023 - Present
Research on LLM evaluation and refusal behaviour across RAG, Text-to-SQL, and multilingual digital libraries within the ITSERR Project.

- Led research on how LLM safety mechanisms and retrieval systems behave across tasks, contexts, and sensitive corpora.

**Visiting Researcher - Text-to-SQL** Athens, Greece
*DARE Lab*, Research Lab April 2025 - July 2025

- Developed evaluation methods for Text-to-SQL systems operating over sensitive schemas and interacting with ambiguous or unanswerable user requests.

**Researcher - ML** Fisciano, Italy
*A.I Tech*, Company 2023
Awarded a Research grant from A.I. Tech srl for the creation of a Machine Learning-based intelligent system for the management of public transportation.

- Played a key role in defining users' requirements
- Refined the data collection and augmentation pipeline
- Led the design and implementation of the intelligent monitoring and predictive system

## RELEVANT PUBLICATIONS

- **COVER: Context-Driven Over-Refusal Verification in LLMs** Findings of ACL 2025
  Led COVER, at the time, the first evaluation of context-driven over-refusal: cases where an LLM rejects a benign task because of the supplied documents rather than the user's request. We studied this behaviour using public Hadith and Talmud corpora, historical Islamic and Jewish sources whose discussions of warfare, law, purity, and personal conduct can activate safety filters despite scholarly intent. The same evidence was far more likely to be refused during translation or summarization than question answering, sometimes for every request. Providing more documents generally reduced refusal, but refusal rose when more unsafe documents were present, showing that both the task and the evidence shape safety behaviour.
- **Text-to-Refused-SQL: A Comprehensive Evaluation of LLMs Refusal in Text-to-SQL** SEBD 2025
  Led the first systematic study of schema-driven over-refusal: cases where a Text-to-SQL model rejects a legitimate request because the database schema contains personal or sensitive fields. We built a benchmark by adding synthetic personal and sensitive fields to diverse healthcare databases, then varied whether questions requested individual or aggregate records, whether example values were shown, the safety instructions, and the user's stated permissions. Behaviour varied sharply across model families: affected models refused even non-sensitive questions because sensitive columns appeared elsewhere in the schema. Most strikingly, stating that the user was authorized more than doubled refusal in almost every affected setting, suggesting that legitimate access information can be misread as an attempt to bypass guardrails.
- **Sensitive Topics Retrieval in Digital Libraries: A Case Study of Hadith Collections** TPDL 2024
  Led the construction of Question-Classify-Retrieve, a framework that generated questions from roughly 4,000 hadith passages, classified each passage as sensitive or non-sensitive, and checked whether retrieval systems returned the source passage. Across about 12,000 questions and ten embedding models, sensitive passages were usually found within the top 25 results but ranked lower than non-sensitive ones, with ranking degradation reaching 26 percent while BM25 remained comparatively stable. This distinction between finding evidence and ranking it prominently showed how modern embeddings can make parts of a digital library less visible to researchers before an LLM generates any answer.
- **ABISS: Evaluating Text-to-SQL Systems Through Agent Interaction** arXiv 2026
  Led ABISS, which unifies previously fragmented reasons why database questions can be ambiguous or impossible to answer, including missing schema elements, missing external or user information, and conflicting evidence. A multi-agent validation pipeline uses several open-source LLMs to generate and check 7,145 questions spanning answerable, ambiguous, and unanswerable cases from arbitrary databases. Unlike benchmarks built from fixed conversations, ABISS tests the complete interaction with simulated users: recognizing the problem, asking for clarification, interpreting the reply, and producing SQL or explaining why no answer is possible. Across BIRD and Spider, agents often recognized that a request was problematic without identifying why. Providing the correct category improved explanations and SQL accuracy, yet models still struggled to turn useful clarification into the correct SQL query.

## SKILLS & TOOLS

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| **Core** | LLM evaluation, safety and over-refusal, agent evaluation, benchmarking |
| **Methods** | Multi-agent pipelines, RAG, Text-to-SQL, OCR/post-OCR, synthetic data |
| **Tools** | Python / PyTorch / Hugging Face / vLLM / TRL / Docker / SQL |

## AWARDS

- Best short paper award at JCDL 2024 with the paper *"Arabic Text Enhancement with GPT for Digital Libraries"*.